

ActInf Textbook Group ~ Cohort 8 ~ Session 5

(Chapter 3) ~ 3/20/2025

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Okay. Hello again, cohort 8. It's March 20th and we're in our first discussion on chapter 3. Lots of ways to enter into this, but just to start, would anyone like to um point to a quote in three or a figure or any area of three or how they see chapter 3 in the context of the book or how they see the low road and the high road? Uh well, I have a question about the sort of the end, I guess. um uh where it says I'll just read the whole paragraph even though my question might not really be about the whole paragraph. Uh an active inference agent appears to perform approximate basian inference under a generative model to maximize evidence for its model that is a self-evident agent. The perspective bit of inference is realized by selecting courses of action or policies that are expected to minimize free energy f in the future. This formalism leads to a novel view of optimal behavior in terms of Hamiltonian principle of least action. Uh yeah, I guess that's I guess my question is about that sentence. Um I'm not sure. Uh yeah, I I guess I don't not sure I believe that. So there's a discrepancy in what you observed and what you believed. And so there's two ways to play that out. By changing what you believe or by taking some kind of action. Do you believe that? Um well maybe there might be more than two in in uh but um there might be those two. But uh so one problem I see with this is I had I had it highlighted. Uh so it says it says something else. Hold on. Uh let me try to find what I was looking for. But yeah, being a little clearer about what what part you Yeah. So the like there there's Yeah. So that was at the end of the chapter to be clear that what I just read was from the very end of chapter three but there is also a tie in uh okay yeah so it gives three associations um between active inference and Hamiltonian dynamics. The first one is just you know a pretty vulgar analogy. It just says that it's a minimization process. Um the second however is where I have more of a problem. It says uh okay I guess I'll read this whole paragraph. Uh the second point of connection between Hamiltonian physics and active inference arises from a more direct association between Hamiltonian and probability

measures. Uh this appears to just be uh more or less um assumed. Uh the the idea here is to associate the conserved Hamiltonian with the energy of the system which of course you know it you the Hamiltonian usually is the the energy of the system at least with what you would deal with. Uh but then it oh wait yeah so here here's the problem. It says that remember that quantities here referred to of energies have had the form of negative log probability. The only reason you're allowed to do that is because of because it's the Boltzman distribution, right? You use the Boltzman distribution to to to have that equation. Um this reflects energy. This reflects an interpretation of energy is simply a measurement of improbability. That is simply not what energy is. energy is simply the conserved quantity that you get because um because the laws of physics are symmetric under time translation, right? And then uh and then it says on this view conservation of energy and probability are equivalent laws. And I have a problem with that too because um the I the idea that conservation of energy and conservation of probability are different actually underly the construction of the Boltzman distribution. So uh if uh oh I should by say by the way this is on page 55. Um the yeah these are great points and questions. I I But you let me finish because the really important point is that it says that uh conservation of energy and probability are the same. But that probably can't be the case because if it was then you wouldn't have the Boltzman distribution because the Boltzman distribution is constructed by assuming that those are two different conservations that you have. If you don't have the Boltzman distribution then you also don't have the association between energies and negative log probabilities. So, uh, yeah. Yeah. Yeah. Great point. Write it down. Just with you've highlighted the part that you've taken a clear perspective on. Write it down. Post it in Discord and/or Zenoto. Let's ask those who can evaluate it in a more methodical way. But great points. Okay, if that's uh if if this topic is discussed, I would like to uh uh question a different figure in the book in the chapter. So if if this is discussed okay then my pro u what I would like to get more intuition on is figure 3.2 two which essentially states that internal uh dynamics will uh go will do the thing uh will not mimic external dynamics but I don't understand how how to interpret this picture and how to like get some real life examples about how how I should really interpret that and 3.2 uh figure okay just to to lead into 3.2 do. So first Nacho asked action states refer to the possible actions the system I do almost. So the possible actions that are available are the affordances. So that's like the the the dictionary of words that are possibly playable. The action state in a given moment or the active state is the actual selected action in that moment. So possible actions would be like I could turn the wheel left, right or straight and then the action state for the markoff blanket of

the car would be the wheel is turning to the left. So that's what it gets interfaced. So yeah, so what gets interfaced to the environment is the selected single action whether it's from a discrete or from a continuous space but from the internal side the deck of cards. Okay. So we're working with this topological partitioning of the agent environment scenario. Blanket states intermediate between internal and external states. It's called the particular partition as sort of a little pun because it's this one particular way to partition it. It's not saying this is the only way to partition it. It's saying this is a particular way to partition action and perception and there's other ways which can be proposed and have been proposed also particular carries this connotation of like carving out a particle. So if we think about brownie in motion and sort of the example of looking at a dust particle vibrating in the microscope in the water from like thermal vibration um that's a particle on a backdrop and so this partition carves out the particle which is like the agents or the thing from that which is not the thing. Okay. Now, how does that look probabilistically? Okay, so we have three different [Music] um variables that are going to be looked at here. This one I always think for this uh diagram, the Girdleerbach book cover. Okay, so we have one object and then it's going to be projected in three different dimensions onto two dimensions. So we have X which is like X marks a spot latent or hidden or external state. So that's like the true temperature in the room. Then there's μ which is the internal states like μ for mind b for blanket vertical line means conditioned upon. So that's what these um different two-dimensional projections at any given moment you have a value for external blanket and internal. So any given moment in the simulation is a point in a three-dimensional joint distribution. Now we're going to condition upon some variables and look at the remaining um scatter in the two dimensions. So here is internal states with respect to with linear regression on blanket states. So this is like this relationship here is external states conditioned on blanket states. So now these two it it these both are sort of like directly connected because external states are connected to the blanket. They have an actual direct causal edge. Internal states are connected to the blanket. They have a direct causal edge. So these two the fact that there's like a statistically significant regression or a mutual information is kind of given by the definition of direct causality. But here we see that there's actually also a probabilistic relationship albeit a little bit more scattered because it's more mediated between internal and external states even though they don't directly connect causally. What this crosshairs represents is that here is x and note that the scale like just look at this you know here's the scale is -105 but it's just heristic but here it's a little different here we have x is at -6 and the internal state is at 48. So this curve is like this hump like imagine if this were scattered

if these points were importance sampling based upon the elevation of a landscape. That's sort of the optimistic view. We're sampling more from the higher probability areas like those are the data points we selected from the experiment. or in flip that picture. Now the lower higher probability things have a lower energy. And so then we're thinking about like it's a valley and we're sampling the location of a ball in the valley and it is going to be found more in the bottom of the valley. But in either case, if you take a cross-section right here, you'll get this distribution. And so what this distribution represents is the belief distribution about external states the expectation of internal states conditioned on the blanket. So this is kind of showing how you have a causal edge between external and blanket, causal edge between blanket and internal and mediated through that chain of causality. You can get mutual information between external and internal states. And then you could have a situation where like the internal states are about temperature. External states are temperature and you have a you know one one or a positive or simple linear correlation. Or you could have a more complex mutual information setting where there still is information of the internal states on the external states but it's not like just trying to track it like a stock price and say I I see that because you speak of this mutual information and that manifests itself in these trends. Um but they also speak about it mimicking the causal structure of the world like internally. Um is that also what they claim here about know the internal states trying mimicking the causal structure of the external world or is that interpretation not justified here? So it's not required. For example, common is external state will be a temperature value that's continuous. And then the homeostat like the agent or the thermometer on the wall is just three discrete states like low, medium or high. Or it could have five like too low, lower boundary just right, upper boundary too high. So theru the structure of the agent's internal generative model that distribution or it could be totally nonsensical like the temperatures from the outside could be a Gaussian distribution and the agent could be trying to do the best variational inference they could on a totally different variational family. So it in the general case it doesn't have to have the same structure. So this comes up a lot in modeling because and why it's so important and chapter six goes into these kinds of points is like what is the point of the model because for example if you have an agent who knows the consequences of its actions like it knows that when it does action up it's going to move up in the world then it is going to do inference on where to move but you've kind of sidestepped this question of how it comes to learn what those affordances or the possibility of those affordants changing their consequence. Or if you have an agent that already knows that it's locked into temperature, then you've kind of given it a causal head start or structural head

start where it's like in the right ballpark. Or another example in the T-maze that comes back in chapter 7, the rat already knows that there's two contexts that there's like two signs that either signal the food is probably on left or the food is probably on the right. So chapter 7 is like saying we've already trained this rat to understand the teammate's game and to undergo this seiosis of learning that the sign at the bottom represents the probability of reward being on either of the two arms of the teammates. So that model doesn't address how do rats learn about how food is on one side of a teammates or not. So I think I understand. So for me uh so for example I am um a system and I have an internal temperature and even though my internal temperature doesn't directly influence the external environment we do see that my internal temperature does is synchronized with the external temperature and that's like then what like this synchronization but also V2 points that we get these kind of synchronizations with this particular partition also motivated almost I might say the slightly simpler case is is the internal belief about outside temperature and then the outside temperature. You could also have a model where there's an outside temperature and then there's a system with an internal temperature and it's doing its own temperature homeostasis that way. It's like th those are those are structurally different models. So, you know, all models are incomplete slash wrong, some are useful. That's kind of the whole point is there's a variety of models and depending on what you're trying to reduce your uncertainty about or frame various combinations. And in practice, we build like kind of portfolios of models and look at similarities and differences in different attributes about them. Not just try to build one and just say, "Okay, this was like the model."

Uh John uh hello. So I have some questions that are related to each other about priors and preferences. Uh so I I guess the first one is um when you talk about priors typically the posterior is used as like the prior for the next phase of inference. So, does the uh do the preferences change a as you do successive sequences of inference? Great question. First, absolutely correct. like the prior if people just say just that alone usually they're talking about the prior on the latent state d on s but technically every variable has a prior then the posterior the post update value becomes like the prior for the next moment that's the basian experience about preferences you can have a model with preference updating you can have a model with preference learning but you don't have do. So you can have a preference that updates in a prior posterior basian fashion or you can just have a fixed preference variable which is corresponding to like an infinite precision fixed preference distribution. Okay. So, so the second thing is I I think I could like imagine a situation in which the prior is different from a preference. Like I could imagine a child who might have a different prior uh to meeting lions.

A different prior probability than a different than uh their preference in the sense that uh the probability might be higher than the preference. Uh and I don't see how those would be the same in that case. Yeah. So it's like for example like I would really love to win this lotto but I'm also reconciling that with I have a low belief that it's likely you know it's not likely to happen. Is that kind of the situation you're talking about? Uh yeah just you know if it's a prior. So um you know you could clearly get around this by saying it's not really a prior in some situations. So, but you know the thing the thing about the child and the lions is that it's a real prior because the child hasn't met a lion before. Yeah. So, let's just say we have preference on seeing lion not seeing lion. And this variable is going to look like like this. We're just taking that that as a binary observation. I'm going to say one and 100. Okay. Massive preference to not see a lion. That's the the preference distribution is over observables. Then latent state, you know, latent state, hidden state, external state, all the same thing. prior on latent state and this is context with lion context no lion and that variable the child is going to say I think it's 80% likely that lions are around 20% likely and then the a matrix would be a 2x two let's just say that it looks like rows and columns are these two. So here with lion always maps to seeing a lion. No lion always maps to no seeing line. And this is the identity matrix. If this were all 0.5s it would be the whole situation would be grayed out. or if it were like 0.9 and 0.1 it would just be like kind of a messy or slightly ambiguous observation. So even within a single layer model so not appealing to like hierarchical like beliefs on beliefs and things like that we already get the exact kind of behavior you're talking about where it's possible for there to be a preference for something to happen and a very different belief about how likely it is to happen. But that's even just within a very simple single layer model. If you want to talk about like kind of, you know, more sophisticated things, those can be built out from there. But it's actually expressive enough to hold that at the base too. So So you say that the uh the connection between prior and preferences isn't um isn't iron. Priors on what? Um, well, we could use the lions, right? Right. These are separate variables. So, they're up to the modeler or the situation of interest. Okay. But I think it says uh I don't Hold on. I have to I have to go find the text. I'm think I think I'm arguing with Hold on. I mean, don't hold on. I I'll be back in a few minutes. Yeah. I'm also here. I'm going to bring in this is kind of a a very very classic slide. This is from live stream 37. the paper um it's called free energy a user's guide. They talk about how the same distribution can play a a sort of linguistic onlogical role as the probability of something occurring. That's sort of the naked statistical sense, the preference as well as the fitness in the historical tally setting. So if we think about this in temperature, it's like there's my body

temperature throughout the day. It's like Gaussian around 37. So that's just the probability distribution. But that also can be reflected as like the constraints or the distribution assuming that's like homeostatic and healthy. And then similarly that can be seen in the in the sort of population scale as being like adaptive to fitness because if there was some other distribution that had higher fitness then that statistical phenotype would have come to predominate in the environment. So is it because of fitness that you the child doesn't grow to not expect lions? Yeah, this is a great question. So then this is this is this is um the kind of work that that um with Bristen and others I've focused on in the last several years too. You know where do those priors come from? Well so what I'm really asking is what's the exact difference between priors and preferences like in the general case? Well, no, an active inference because it seems like they're the claim is that they're pretty much the same. It seems to say that in the textbook. Which part does are you looking at in the textbook? I I don't know. I have to I have to look for it, but it seems to say it in chapter three and in other chapters. Yeah. Like the general case is very general. If you look at a specific model or if you make a specific model then the difference between this and that will just become so plainly deflationarily clear. It's like I mean are subway map are subways drawn thicker or thinner than highways on the map in general? It's like well in general no because you could do you could have a map with or without subways or you could draw them any way you want. But then you look at a specific map, you go, "Okay, it's thinner." So it's like some of these things in the general case, not like it's the closure of the discussion, but in the general case, you go, "Yeah, you could do it one way or you could do it another way because it's like the general toolkit." You know, how many wheels does a wheeled object have in the general case? 1 2 3 4 5. How many wheels does this car have? Oh, four. Oh, it has three. Okay. But that's like the literal difference between the general case and the framework and the toolkit and the possibilities and the modeler degrees of freedom and the specific generative models that you build, which is to say the specific or the particular joint distributions defining the generative model. Okay, another question from not show. In this sense, active inference defends a form of modal representationalism similar to Fodor's idea, but modal, not just digital. Do you want to like add a little more context, Nacho, or does anyone have any thoughts on this? Certainly this question of representation has come up like just looking at the titles from streams and papers. questions about representation in semantics. How do signs and signifiers and represent uh come to represent and be understood to represent and you know reference questions like that? this um paper of Matt Sims. This one is kind of this one's kind of a classic to this earlier point where it's like there's the general

setting where it's like you know wheeled objects could have any number of wheels and then there's the specific settings. What Sims does in this paper is says okay on four different ways of thinking about the concept of representation organizational representation structural content and functional representation. There's a representationalist and a non or anti-representationalist view. And then the paper is what does it mean about our own understandings in where we come down on representationalism and not on these columns. Representationalism means that our experience is uh one big representation of the outside world or like it's how I see this list that our experience is a experience explanation of our sens sensory input is that what representation doesn't mean you're you're in the ballpark like each of these four I just don't really remember the details. Each of those four different columns take a different view on what representation is. But that's a great paper to look at. Oh yeah, I'm learn. Yeah. like does something represent does cup represent that cup if it um propositionally invokes the cup but I don't actionably know what to do what about the other setting where I have no propositional understanding of it but I go to grab that cup when somebody says cup and so each of those or what you know and so And it relates to this earlier discussion on temperature. Like what does it mean for the agent to have a representation of temperature of the outside temperature? Does that mean that they have a variable that needs to have mutual information and monotonically track it? But what if it's a terrible representation? You know, is a bad representation still a representation? What if it's actionable but has a very different semantics? John Uh okay, I found what I'm was complaining about in the textbook. So it says uh let me make sure this is the right. Yeah, it says it unites. So in contrast in active inference, behavior is the result of inference and optimization is and its optimization is a function of beliefs. This formulation unites notions of prior belief and preference as discussed above using the notion of expected free energy. And then it says something I think about the uh optimistic view. Yeah, I'm sure there's a ton of ways to read this, but it unites them like if you unite flour and water to make dough. So here there's the uniting of these two different variables into a joint distribution. So it's not saying that they are pointing to the same variable. It's saying we're uniting two different kinds of things, prior beliefs and preferences. You know, D on S for example, and C. uniting different variables in a joint distribution in order to get to this unified type of model or map for sensemaking and action selection systems. Okay, but it says it's endowing the agent with an implicit prior belief that it will realize its preferences. So that's that's like saying it endows the child with an implicit prior belief that it will uh realize not seeing any lions. Using the notion of expected free energy amounts to endowing the agent. So if we don't have an action selector,

we just have an a like a rock that's not engaging in counterfactual policy inference. It's not saying the EF is like expected in the pragmatic uh expected pragmatic and epistemic value equation 2.6 in the textbook. So if it is engaging in this energybased method with an expected free energy functional such that pragmatic value which is to say alignment with preferences are seen as more likely that's why we sample from them. Okay. So when you say energy I just have to clarify because it says in the textbook that energy is seen simply as a measure of improbability. Is that how you're using energy? Yeah exactly. So here there's there's a direct we have the policy prior π then expected free energy operates on the policy prior and it updates it according to pragmatic and epistemic value. So let's just leave epistemic value off to the side um for a second. So here's figure 2.6. So here's expected free energy on policy. We have a vector of policies. Let's just say we could go up, down, left, right. our policy prior. They're all even. Or you could have a habit, but they're all even. Then we're going to update that probability distribution over action according to the epistemic and the pragmatic value of those actions. So in that situation where we are explicitly upweighting the probability of selecting actions that fulfill our preferences then we have endowed the agent with a prior belief that it will realize those preferences. If you said, "I'm going to make a counter ef it seems also in other chapters such as chapter 2 that it talks about uh preferences being defined by priors." And I'm just not sure that that can be made to work in all situations. Yeah. Find that situation and then you've made an advance in FEP. Well, well, it's the lion thing with the children, right? Cuz um what does the prior of the the children with the lions have to do with their preferences? Okay. So again, we're in a thought thought kind of counterfactual where the only observation is whether we see a lion or not. And then the latent state is whether we're in a context with lions or not. And so it's like what what is the aboutness of what is the relationship between the preference for seeing it or not and the belief about whether they are out there or not? Well, that's sort of my question. Yeah. What is it that connects the dots? Lions. Children in situations with or without lions connects the dots. It's like, I made this map. What is it that connects all of these representations on the map? It's like territory. But if it's a fantastical everchanging territory, don't expect the map to have too much coherence. But if you pick a fixed scenario for what is the actual process that you're modeling, then you'll find you can make many coherent maps with not much ado. So, so there's a difference between the prior and the preference in this example, right? Like look look at the the array is there. Yeah. The slight ambiguity is sometimes when people just say a prior they mean a a prior on the latent state like a prior on S D on S . Like if we look at the figures sometimes this one right here three

or D is called the prior. In a more general Bayesian updating context every variable has a prior. Now, some of them are learnable or not. First question about whether the preferences update when the priors update. It could be you can do preference learning or not. Well, I I I just I I just don't understand exactly what the connection between the priors and preferences is supposed to be because in the lion example, they seem separated. Um here's the context. Is there are we in the lion context or not? Here's observations about lion or not. Preferences have the same form as a distribution over observations. The reason why models always have almost always have updating of beliefs on latent states is because that's where actions intervene. So if there was no updating of latent states, there would be no purpose in intervening. So action selection would be this moot kind of epistemological question. So for the agent to believe it has agency, which may not be the case in the simulation as you set it up, it could believe that it's changing the stock market just by like switching channels on the TV. So it doesn't need to have a veridical representation, but that's that whole representation question. If you have a representation that's nonveridical, is it a representation at all? And the general case is read the Mad Sims paper, see what you think about that and where you put yourself. But in the particular case of a given model, you resolve that just by writing out the distribution you're talking about. Are you talking about a person regulating their temperature and their air conditioner and the heater work or are you talking about a situation where the air conditioner and the heater are not plugged in, but they still believe that they are. But those are modeling different scenarios. But even just right there, that's why the general case doesn't come down on such an opinionated way because you could just think of those situations right off the bat where the agent does or doesn't have causal efficacy. So if the agent is the causal efficacy the thing that tells you whether the priors are the same as the preferences or not? Okay, this variable priors on latent states is not preferences is never the same as the preferences. Okay, it can be related to the same situation. I can have a preference on body temperature like observations, interceptive temperature observations and prior belief on temperature in the room. They're separate variables. So, I feel like the textbook argues in several places that they're the same. Um, the textbook is just pixels. Your perception of it is what you're experiencing. Uh, okay. I'll look through it and I'll probably send you some quotes because I feel like this is a thing that it says and it seems like a thing that you're saying half the time. Uh, but yeah, great. Absolutely. Not saying I'm correct, not saying the textbook is fully correct, but it's recorded and it's written. So, pull out the specific parts and then that's the conversation. Well, I think it's in chapter 2. So, I found this one part um uh but I think it goes

on. So, yeah, it says that it says that the agent will have an implicit prior belief that it will realize its preferences. So does the child have an implicit prior belief that it will realize it's okay using the notion of expected free energy amounts to endowing the agent. So okay now we're going to go back to the child. Now the child the policy okay we have run towards lion forest is option one and then run away from lion forest. So now we're bringing action into the picture. Okay, if it just Okay, so now its preference is not to see lions. So now let's just say that it's its habit policy prior. Again, this is like why we use the ontology like this because it's not in this infinite set of words like there's a policy prior it let's just say it's 0.5 then it's going to do expected free energy calculation or we're going to model it as if it's doing expected free energy calculation the policy and and we're we're going to discount epistemic value. Okay, so we're just looking at this setting with just pragmatic value. Maybe curiosity makes it go there, right? That's that's the second question, but just on the pragmatic preference satisficing elements. Okay, which one of these policies is going to have higher pragmatic value? Well, running towards the lion forest makes me believe that we're going to be in a state where there is lion, which makes me believe we're going to see a lion. Running away from I'll transition into a state where there's no lion, which is more apt to have the not seeing of a lion. So, this one is going to have higher pragmatic value. Yeah, but it won't have enough value to change that 82 distribution into that one. Like really, you should look at the specifics of models because you could have it so that it's so overpowering that it updates from 991 to 199. That's the whole situation with the frog is here. It's 10 10% it's a frog and then after it goes to 0.9. So that is belief weighted updating which is where if you didn't pay attention to the observation it has no impact. You carry your prior forward. If you had full attention to the observation you would say okay well now it's 100% it's a frog or it's 99% it's a a a frog. And then everything in between is the the gradation of attention and that's attention weighted basian belief updating that just gets down to the quantitative detail here. I'm just laying out the structure of it. Okay. So, so is the the short answer that the prior is different from the preferences and that these are just different numbers, right? The short answer is the prior is different than the preferences. They're two prior on latent states is over here. Three preferences comes into play over here. We can look at 7 uh three. D is a prior on latent states. C is a probability distribution on observations. They don't even have the same structure. They're two separate variables. They come together. They're united in this model where if you use expected free energy, it implicitly endows the agent with this belief just focusing on the pragmatic value that it is taking actions that make its preferences more likely to be realized. Secondary things

like how does epistemic value modify that picture? How well does it modify um how how well does it achieve its preferences? you know, how how agentic is it? The there's all these secondary questions, but just to the prima facie and all of the quotes that you've raised, I hope that makes it kind of clear and it becomes clear when you actually write out a specific example because the general case delightfully doesn't resolve it. Anyone who hasn't um raised a question or said anything want to just like add any pieces some interesting part on three or another figure or quote Hey, Daniel. Uh, this is Jerry. not uh not necessarily specifically related to that, but I did have a question um kind of cuz you brought up that u that article about like Evo DVO. Is there anything tying this um this framework to more of an evolutionary like predatory prey type of view of things in terms of human behavior that comes into play like as to how we you know search for patterns and consume patterns if you will out there as you were referring to like pixels but then we make meaning of Yeah. Few few thoughts there. So one interesting connection with a predator prey that will come up in chapter 8 in the continuous time generative model is actually the way that lock goer models are used to to model winnerless competitions. Mhm. So here's like the the true classical ecological here's a threepart ecosystem where you know and there's there's parameterizations that lead to collapse of populations etc. But this is just one setting where you have these three cyclic um you know these model and then also here's those those same the same GEB book cover here's this three-dimensional phase space being projected onto two 2D phase spaces just like we saw with internal blanket external but here with yeah um so that has been used in SPM and neuroimaging to model relations among activity in brain regions with like certain brain regions that have positive edges or negative edges. But even when there's a positive edge, it might lead to increased activity and another positive edge and a negative edge. So like the plus plus minus triad leads to dampening feedback with a lag. So it's been used extensively in dynamic causal modeling and continuous time modeling. Now more to the human side. Maybe look at the epistemic communities. Yeah, I've seen that one before with Mao. Yeah, cool. So that's kind of like that. Um I don't know if anyone has done like lock voltera too much further on some specific um but but you know like right that those are the seeds of pursuing that inquiry you know okay would you be measuring this could be you know the sentiment score in three different cryptocurrency chains or this could be you know three politicians and their polling numbers or it sentiment of this or that and then it's like these are kind of the macroscopic dynamic causal model that this is basically giving the toolkit around. Got it. Yeah, that's where my head was going because I work in marketing and communications and like you know

we've got the purchase funnel which is just a bunch of madeup stuff. But I was trying to think of it in terms of more of that um that predator prey model if you would and like getting some uh bringing some computation in around like how those steps are for a predator to go and hunt and consume if you would because I think there's some connection there in human behavior that we haven't quite made computational in my world. We tend to do it more like it's folk wisdom. But I think there's parallels and I think it would explain a lot. But it's good to know I'm at least pointed in the right direction. I did find one computational model of uh predator behavior that I think was kind of interesting, but I was just curious from your perspective if there was anything else. But it looks like I'm at least in the right ballpark. Yeah. Yeah. like predatory lending or things like I mean the just where is that qualitative metaphor coarse graining some more actionable specific quantitative models that could be I think where it's interesting like you said it's it's it's a different levels right like I think consumers it's funny we call them that right but like consumers um they are hunting if you will for something that would effectively um make them feel better. But to your point as well, corporations like basically I think put out like a lure, if you will, like they're they're affordances they're putting out there to be picked up like you know, some in a more symbiotic way, some in a more parasitic way. So, I think there's definitely some uh biological uh research that we could bring in to like give that a better metaphor. Yeah. And these kind of like nomadic metaphors like where are we looking more at message passing and belief sharing and updating in a multi-agentic model? Where are we looking more at macroscopic macroeconomic summary fluctuations? And then you know where do you want a fusion model where there's like a macroscopic lending rate that's like the temperature and the niche and then there's specific offers and then there's like agent level cognitive patterns you know act now to lock in this rate and then what kinds of cognitive structures have what kinds of responses and updating of their habits. Yeah. Yeah. And um I think it's a good call out on that locker because I'd seen that before and I think it's tied into some economic models if I'm not mistaken. So um it that's an interesting place for me to dig in a little bit more. So thank you for that. Yeah. I've I've also seen some like like ecology of fear. Oh, okay. This is kind of an other than being kind of a intense sounding concept, but it's just like predators and prey, whatever, however you want to think of it, food chain, food ladder, food web, you know, trophic, they're cognitive systems. So their biosemiosis and their cognitive processes, it's it's not just about like the last mile of the cheetah chasing the gazelle. They're both engaging in these cognitive evolutionary strategies where over success might reduce population fitness. And those are kind of higher order

evolutionary consequences that intergenerational frameworks could start to model because what if you what if what if local depopulation of deer isn't high fitness for populations of carnivores? Mhm. But but then at the single agent level it's like well optimal foraging says you should just be like on this killing rampage. Right. Right. And I think there there's definitely some parallels there in the work that I do. And um the um the active inference communities work that you pulled up is also very interesting because I think it gets to from our standpoint like the messaging that we're putting out for people to consume. Share voice is something we talk about a lot which would you know inform those priors, right? um and and update the you know update people's beliefs based on what they're what they are seeing the most of and that's what we see in in our work and so that's what I'm trying to kind of noodle on is how do I you know merge that from a communication standpoint um and use some of these I think easier to understand evolutionary terms for people Yeah. Yeah. But the economy of advertising advertising where um is the arbit info arbitrage among other things between how do we spend to update beliefs? Beliefs about what? That's the representation question. Beliefs for whom? Like what other beliefs are tethered or already preassociated with it? Like advertising is cognitive, economics is cognitive. That's like the cognitive term in everything. And then there's the pragmatic turn in the cognitive sciences which is enacted extended you pragmatic all those features and then where all those are sort of swirling is is the space of actionoriented cognitive modeling. Right. Right. Yeah. totally makes sense. There's really some interesting work in my field around uh aggregate attention because there we're getting more to the space where we can not just measure viewability but actual attention paid to messages and that that aggregate amount across channels like really is predictive of how belief changes about the organization that is presenting it and the sales that they then eventually have. They seem very fixed, but we do know that that's a a big thing. Yeah, the attention economy stuff for sure. Um is definitely something that plays into that. Cool. Yep. And then just walking these domains along the road of applicability and rigor from like intuitive, qualitative, metaphorical on but then there's so many terminal branches. There's so many generative models that could be made. So then that's the sort of forwards backwards process in feedback with domain experts and all of this. You know what are the observables? What are the latent states we should be making? What are the preferences? What are the priors on what that we're modeling? What would be a question we'd want a computational model to address? And that's the kind of chapter six questions. Yep. Cool. Yeah, thank you for that. I do have to to jump, but that was super helpful and gave me a few more things to chew on. Fun. Um, okay. Well, so next week if

people want to join at this same time we'll be discussing chapter 8 which we just peaked at here. Otherwise in two weeks at this time um we will be having our first discussion on chapter 4 which is where we actually get introduced to the active inference generative So thank you all. See you next time. Bye.